

Car Showroom Management System

Entities and Attributes:

1. 1. Car

- Primary key: CarID
- Attributes: Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status

2. 2. Customer

- Primary key: CustomerID
- Attributes: Name, Phone, Email, Address

3. 3. Salesperson

- Primary key: SalespersonID
- Attributes: Name, Phone, Email, HireDate, Position, Salary

4. 4. Sale

- Primary key: SaleID
- Attributes: DateOfSale, FinalPrice, PaymentMethod, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)

5. 5. TestDrive

- Primary key: TestDriveID
- Attributes: ScheduledDate, Duration, Feedback, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)

6. 6. Showroom

- Primary key: ShowroomID
- Attributes: Name, Location, Phone, ManagerName

7. 7. Inventory

- Primary key: InventoryID
- Attributes: ItemName, Quantity, Category, ShowroomID (Foreign Key)

8. 8. Staff

- Primary key: StaffID
- Attributes: Name, Role, ShowroomID (Foreign Key), Contact

9. 9. Payroll

- Primary key: PayID
- Attributes: Salary, PayDate, StaffID (Foreign Key)

10. 10. Administration

- Primary key: AdminName
- Attributes: Contact

11. 11. Insurance

- Primary key: InsuranceID
- Attributes: CarID (Foreign Key), CustomerID (Foreign Key), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount

12. 12. Booking

- Primary key: BookingID
- Attributes: CustomerID (Foreign Key), CarID (Foreign Key), BookingDate, BookingStatus, AdvanceAmount

13. 13. Warranty

- Primary key: WarrantyID
- Attributes: CarID (Foreign Key), WarrantyStartDate, WarrantyEndDate, CoverageDetails

System Overview:

The Car Showroom Management System helps manage showroom operations efficiently by automating core tasks such as customer management, inventory handling, and sales tracking.

Key Features of the System:

- Maintains detailed records of customers and their interactions with the showroom.
- Manages car inventory with attributes like model, price, availability, and specifications.

- Processes sales transactions, including customer details, payment methods, and delivery schedules.
- Generates reports based on sales, inventory, and other key metrics to support strategic decision-making.

Database Design:

- Developed using both top-down and bottom-up design approaches.
- Includes an Entity Relationship Diagram (ERD) to visually represent relationships between entities such as Customers, Cars, Sales, and Staff.
- Tables are defined with attributes, constraints, and relational keys to ensure consistency and data integrity.

Advanced Functionalities:

- Custom Views: Tailored views for streamlined access to critical information, such as available cars, high-performing sales staff, or top customers.
- Query-Based Reports: Dynamic queries enable insights like total monthly sales, customer purchasing behavior, car availability, and overall showroom performance.

Purpose:

The system is designed to simplify showroom workflows, enable precise record-keeping, and enhance operational efficiency — ultimately leading to better customer service, informed business decisions, and sustainable growth for car dealerships.